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A Project Report

On

Development of Biosensor Detection System

BY

Kartikeya Srivastava

2023A3PS0385H

Under the supervision of

Prof. Sanket Goel

SUBMITTED IN FULLFILLMENT OF THE REQUIREMENTS OF

EEE F366: LABORATORY PROJECT



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI (RAJASTHAN)

HYDERABAD CAMPUS

(Sem 2 2025-26)

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Special thanks to Prof. Sanket Goel, Prof BVVSN Prabhakar Rao and Prof Arindam Kushagra whose help, suggestions, and encouragement helped us throughout the project. I also sincerely thank them for taking time out of their busy schedule to help us tackle and solve the problems we faced during the project.

Birla Institute of Technology and Science-Pilani,

Hyderabad Campus

Certificate

This is to certify that the project report entitled "**Development of Biosensor Detection System**" submitted by Mr. Kartikeya Srivastava (ID No. 2023A3PS0385H) in fulfillment of the requirements of the course EEE F366, Laboratory Project Course, embodies the work done by him under my supervision and guidance.

Date: 27 April, 2026

(Prof. Sanket Goel)

BITS- Pilani, Hyderabad Campus

ABSTRACT

This project aims to present an overview of advancements in the design and implementation of potentiostat systems for electrochemical sensing applications. We have discussed the design and applications of a NanoStat, which is a type of potentiostat used for electrochemical experiments.

The NanoStat almost achieves full electrochemical functionality, including cyclic voltammetry, chronoamperometry, and pulse techniques etc., using only two integrated circuits: an ESP32 microcontroller with onboard Wi-Fi and an Analog Devices LMP91000 analog front end. The system, with its web-hosted user interface and self-contained firmware, will help enable real-time electrochemical analysis without external software, supporting wireless operation and remote data access at an ultra-low cost.

To improve accuracy and stability, a lot of hardware changes were made. A buffer op-amp with a MOSFET driver was added to better control the counter electrode. A low-noise amplifier with adjustable gain was used to measure small currents more precisely, and JFET isolation was added to reduce leakage. Other improvements, such as signal filtering, stable voltage regulation, and a high-resolution ADC, will help reduce noise and drift. These upgrades will improve the system's accuracy, reliability, and ability to perform real-time electrochemical analysis over Wi-Fi.

In this work, we also look at how electrode geometry affects electric field distribution for gas-sensing applications. We designed and modeled three electrode structures: Interdigitated, Hilbert-Piano Curve A, and Hilbert-Piano Curve B. They all have the same material properties, trace width, and operating voltage. We ran electrostatic simulations using COMSOL Multiphysics by applying a fixed potential to one electrode and grounding the other. The surrounding area was treated as an insulating

air domain. A refined boundary mesh captured the strong electric field gradients that occur at sharp corners and narrow gaps in fractal geometries.

We extracted the maximum electric field strength from each configuration to compare their field enhancement capabilities. The results show that the Hilbert structures have a much stronger local electric field concentration compared to the conventional interdigital electrodes. This is due to their greater number of sharp turns and narrower spacing. Among the designs we tested, Hilbert-Piano Curve A achieved the highest field intensity, followed by Hilbert-Piano Curve B and the Interdigitated electrode. These findings suggest that fractal-based electrodes can improve charge interaction with target gas molecules, thereby enabling higher sensitivity and enhanced performance in sensor devices.

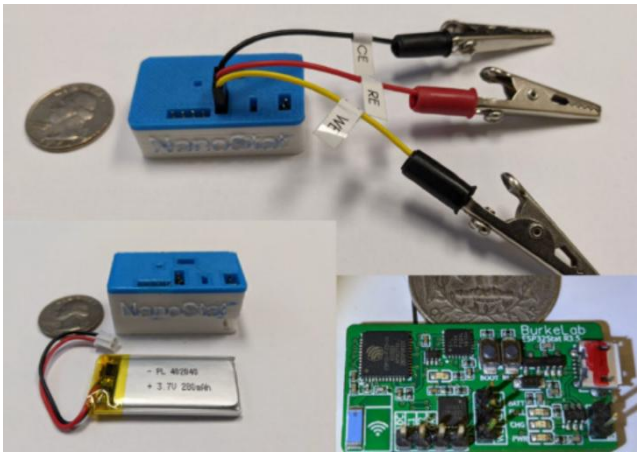
Electrochemical sensing requires precise control of electrode potential and accurate measurement of small currents generated by redox reactions. In this work, a multi-channel potentiostat system was also designed and analysed using LTspice simulation. The circuit uses operational amplifier based control loops which maintain a constant potential difference between the working and reference electrodes while measuring the resulting current through a transimpedance amplifier. An equivalent electrochemical interface was made using a Randles circuit consisting of solution resistance, charge transfer resistance, and double layer capacitance.

This work also presents the design and simulation of a two-channel potentiostat using LTspice for electrochemical measurements. The system also includes a summing amplifier for control voltage generation, control amplifiers to maintain the desired electrode potential, and transimpedance amplifiers for current measurement. Each channel is made using a Randles equivalent circuit to represent electrochemical behavior.

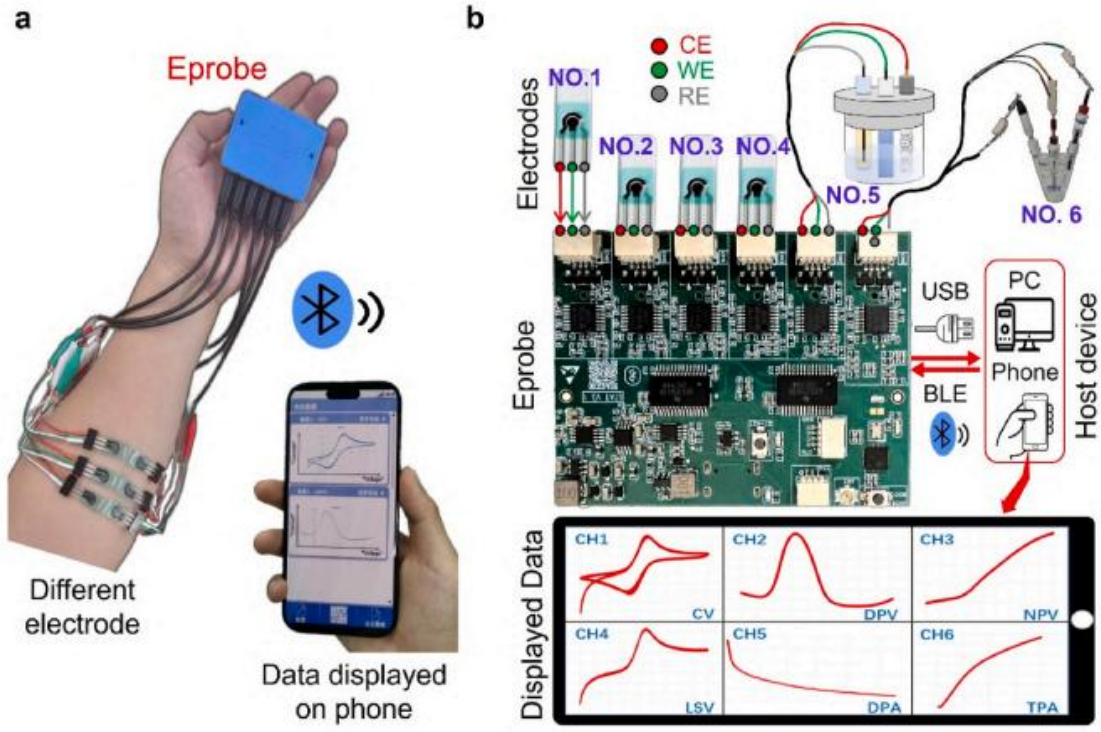
CHAPTER 1 - Introduction

Electrochemical gas sensors find numerous applications in environmental monitoring, industrial safety, biomedical diagnostics, and intelligent automation. The efficiency of gas sensors highly depends on the electrode interface design which facilitates electrochemical reactions and charge transfer. The enhancement of the local electric field intensity at the electrode surface may improve the interaction of charge carriers with target analyte molecules leading to increased sensor sensitivity and reduced detection limit. Recently, miniature electrode interfaces such as fractal and interdigital were explored due to their ability to provide high electric field density without additional volume. Here, the application of NanoStat as an electrode for advanced electrochemical sensors should be considered.

To briefly introduce NanoStat, it is a miniature and cost-effective wireless potentiostat device intended for electrochemical measurements. The key features of this device are its simplicity, as it contains only two chips, an ESP32 module for Wi-Fi communication and digital operations and an LMP91000 chip for accurate analog data acquisition. This open-source device enables efficient sensing and wireless data transmission, making it an interesting candidate for a range of sensing applications in the future. A network of interconnected NanoStat devices may be used to achieve continuous environmental monitoring, personal medical diagnosis and IoT-based systems. By combining NanoStat with high-performance electrode geometries, the sensing capabilities of this system can be significantly improved, making it an ideal platform for next-generation electrochemical gas detection.



Eprobe is a very compact multi-channel electrochemical sensing platform which is designed for simultaneous detection of multiple analytes using electrochemical techniques. It helps integrate several independent potentiostat channels within a single system, enabling parallel monitoring of different sensors.



CHAPTER 2 - About NanoStat

The NanoStat is an open-source potentiostat developed to make modern electrochemical analysis accessible and affordable.

Basic Working Principle of NanoStat

The operation of NanoStat is based on the fundamentals of a potentiostat, which is a circuit device applied in electrochemical measurements and reactions. NanoStat uses a three-electrode system which comprises a working electrode (WE), a reference electrode (RE), and a counter electrode (CE) inserted into the electrochemical cell containing the sample solution.

Potentiostat regulates the potential difference between the WE and RE while measuring the current flow between the WE and CE. Current measured by the potentiostat is a direct consequence of electrochemical reactions taking place on the electrode surface. The analysis of the relation between current and voltage leads to the determination of concentration of the analyte in the solution (such as gases, ions, and biomolecules).

In brief, NanoStat uses potential regulation and current measurements for electrochemical investigations.

Some Relevant Formulae that we used –

$V_{WE-RE} = V_{control}$

$V_{out} = -I_{WE} \times R_f$

$I_{WE} = V_{out} / R_f$

$i_p = (2.69 \times 10^5) n^{3/2} A D^{1/2} C v^{1/2}$ Randles–Sevcik Equation (for Cyclic Voltammetry)

Major Properties of NanoStat

1. **Wireless Operation:**

   It functions totally through Wi-Fi because there's no need for any cables or other software. The data can be accessed and manipulated using any internet browser.

2. **Compact and Portable:**

   It has extremely small size, which means that it can be powered using a tiny battery for hours.

3. **Open-Source Design:**

   All hardware components, firmware, and software code are free for anyone to change, duplicate, or enhance the system.

4. **Web-Based Interface:**

   User interfaces are entirely hosted on the hardware itself through HTML5 and JavaScript, making it possible to control and visualize the process via browsers.

5. **Multi-Technique Capability:**

   Can perform many different electrochemical tests, including CV, CA, and SWV.

CHAPTER 3 – About Fractal Electrode Designs

Fractal electrode designs, such as Hilbert curves, maximize electric field concentration and active surface area within a compact footprint to enhance sensing performance.

Basic Working Principle of Fractal Electrode Designs

A fractal electrode works on the principle of increasing the electric field and the active surface area using self-similar geometry with many sharp corners and tight spaces. When the electrodes are placed under a certain voltage, there is an intensive electric field at the boundary due to its shape. This results in increased charge transfer and interactions between the charge transfer and gas molecules. This makes the use of fractals, such as Hilbert curve, for sensing purposes more effective because it senses better than linear electrodes. **Hot spots** in fractal electrodes arise due to the intensive electric field at the sharp corner and tight space.

Some Relevant Formulae that we used –

$E = -\nabla V$

$|E|=(Ex^2+Ey^2+Ez^2)^{1/2}$

COMSOL calculates |E| directly with the help of the following expression:
|E|="es.normE"

while comparing different shapes of electrodes, the maximum electric field within the sensor area is determined: $E_{max} = \max(|E|)$

Major Properties of Fractal Electrode Designs

Fractal electrodes possess several advantageous properties that contribute to improved electrochemical sensing performance:

1. **High Effective Surface Area:**
   This property of space filling by fractals leads to an increase in the electrode perimeter and reaction interface.
2. **Strong Electric Field Enhancement:**

Many acute angles and segments close together produce strong localized electric fields, enhancing the transport of charges and the adsorption of gas molecules.

3. **Compact Geometry Utilization:**
   Fractals can effectively cover two-dimensional space, making them suitable for highly effective miniature gas sensors.
4. **Improved Sensitivity and Response:**
   This is because of the enhanced interactions between the surface and the gas molecules, along with stronger localized electric fields.

CHAPTER 4 – About Eprobe

Eprobe is a multi-channel electrochemical detection platform based on potentiostats. The main function of Eprobe is controlling the voltage of the electrochemical cell and measuring the output current generated due to reactions that occur on the surface of the electrode. Each channel in Eprobe works as a separate potentiostat.

Basic Working Principle of Eprobe

$V_{WE} - V_{RE} = V_{set}$

- V_{set}= applied potential (input waveform)
- V_{WE}= working electrode potential
- V_{RE}= reference electrode potential

Some Relevant Formulae we used –

$V_{out} = -I_{cell} \times R_f$

- V_{out}= amplifier output voltage
- I_{cell}= electrochemical current
- R_f= feedback resistor of the TIA

$I_{cell} = -V_{out}/R_f$

$I_C = C_{dl}\, dV/dt$

Major Properties of Eprobe

1. **Multi-Channel Capability**

   The design of Eprobe is made possible by using a number of potentiostat channels, which help monitor several electrochemical sensors simultaneously.

2. **Multi-Technique Support**

   The device is able to carry out various electrochemical measurement techniques including cyclic voltammetry (CV), linear sweep voltammetry (LSV), and chronoamperometry (CA).

3. **High Sensitivity for Low Currents**

   The ability to detect extremely low electrochemical currents makes it applicable for the use in biosensors and chemical sensors. It is able to measure very low currents ranging in nano-amperes and micro amperes, due to the use of transimpedance amplifiers. This enables the use of Eprobes in the field of biosensing and chemical sensing devices.

4. **Compact and Integrated Design**

   It is possible to incorporate several test channels into one system by utilizing a single hardware platform, leading to a reduction in hardware volume.

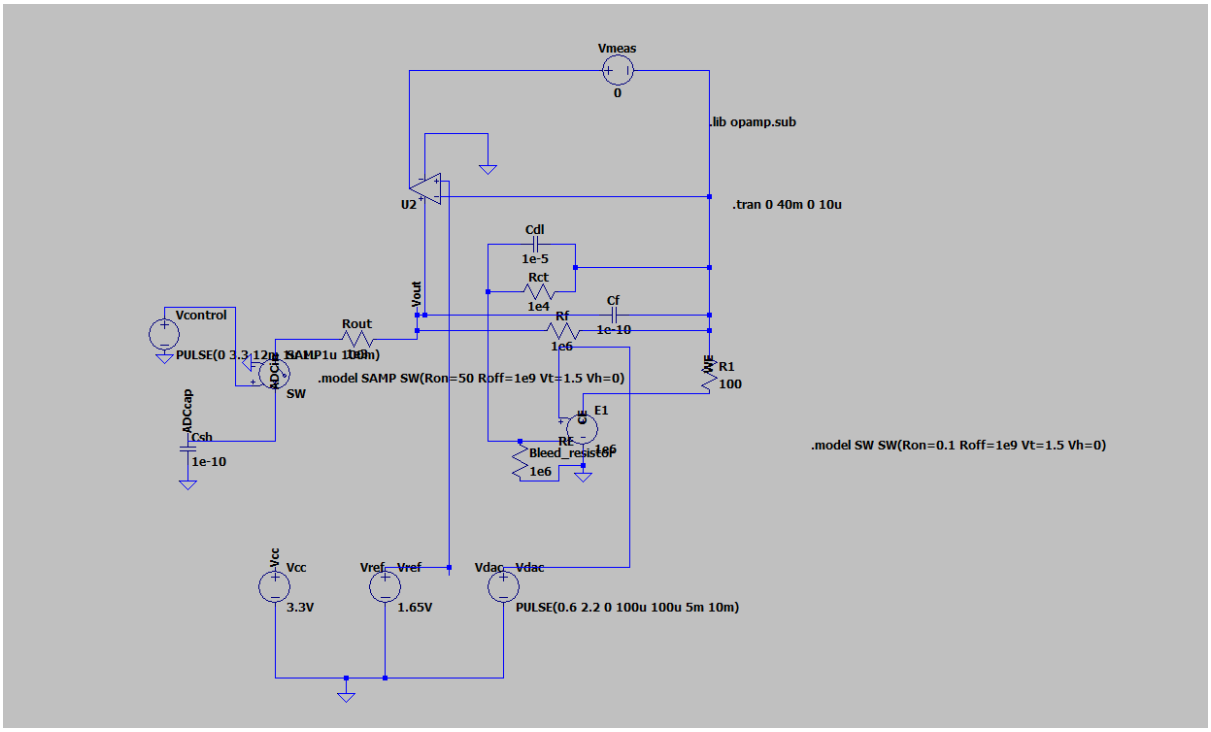
5. **Independent Channel Operation**

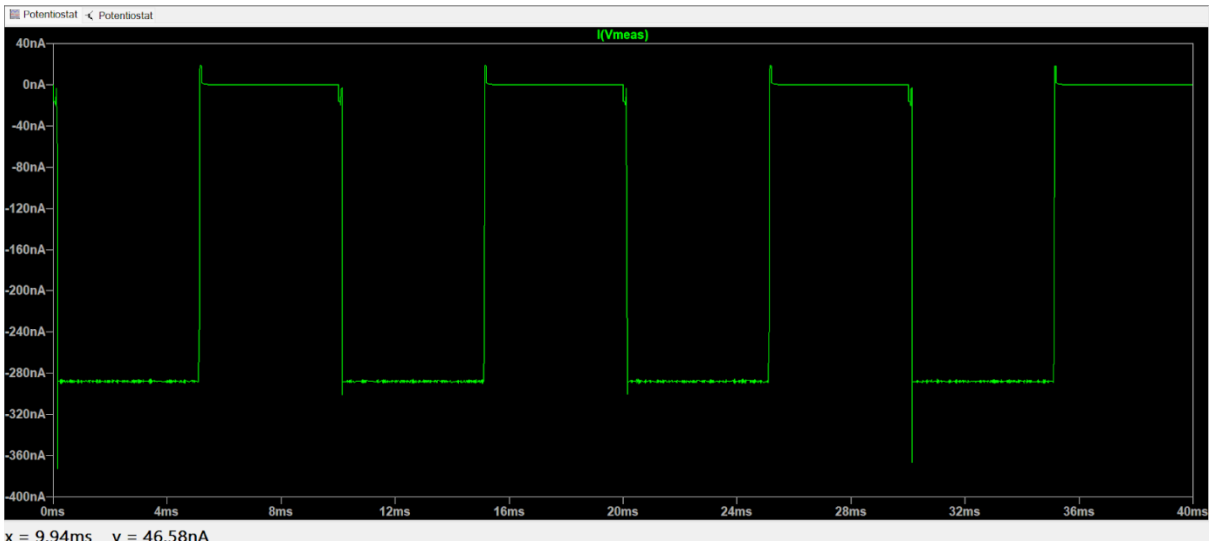
   Each channel operates independently, ensuring that signals from one sensor do not significantly affect the measurements of another sensor.

CHAPTER 5 - Progress Update

For Potentiostat -

We started the Potentiostat design in EasyEDA which is a web based electronic design tool suite. The problem which we then faced was that it did not have in-built working ESP32 and LMP91000. Only the symbolic representation was there. Thus, we decided to shift to LTspice to design the circuit for which we had to start from scratch to design an ESP32 and LMP91000. The first design which we designed was as shown below.

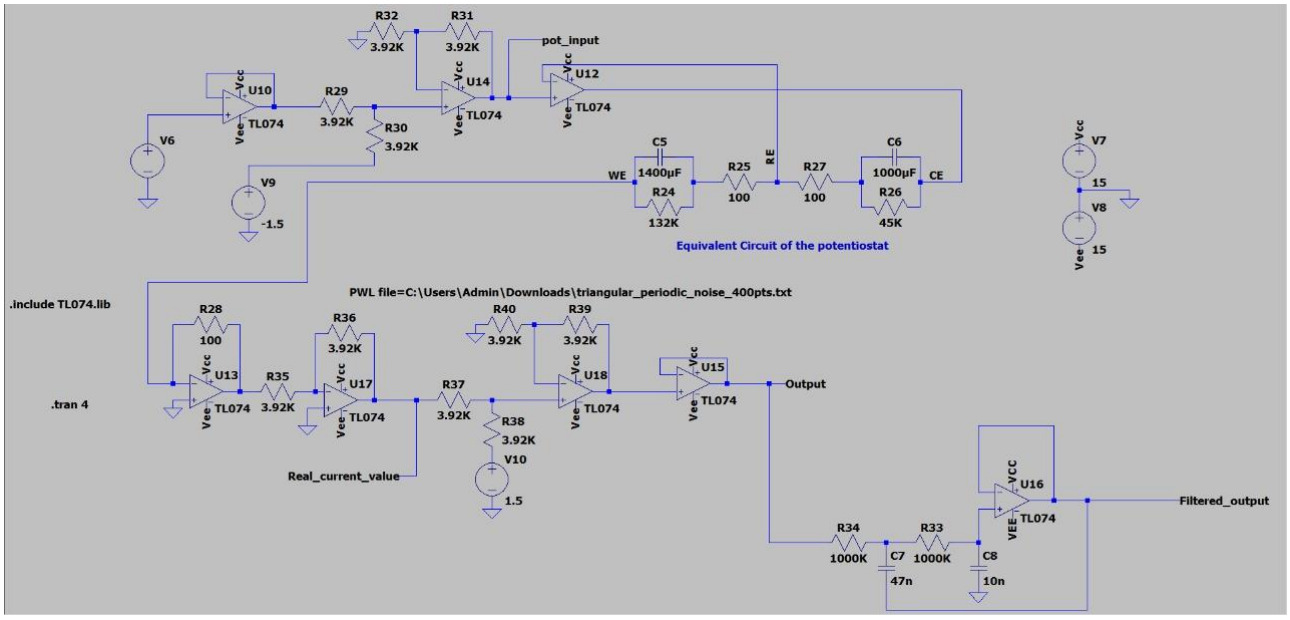




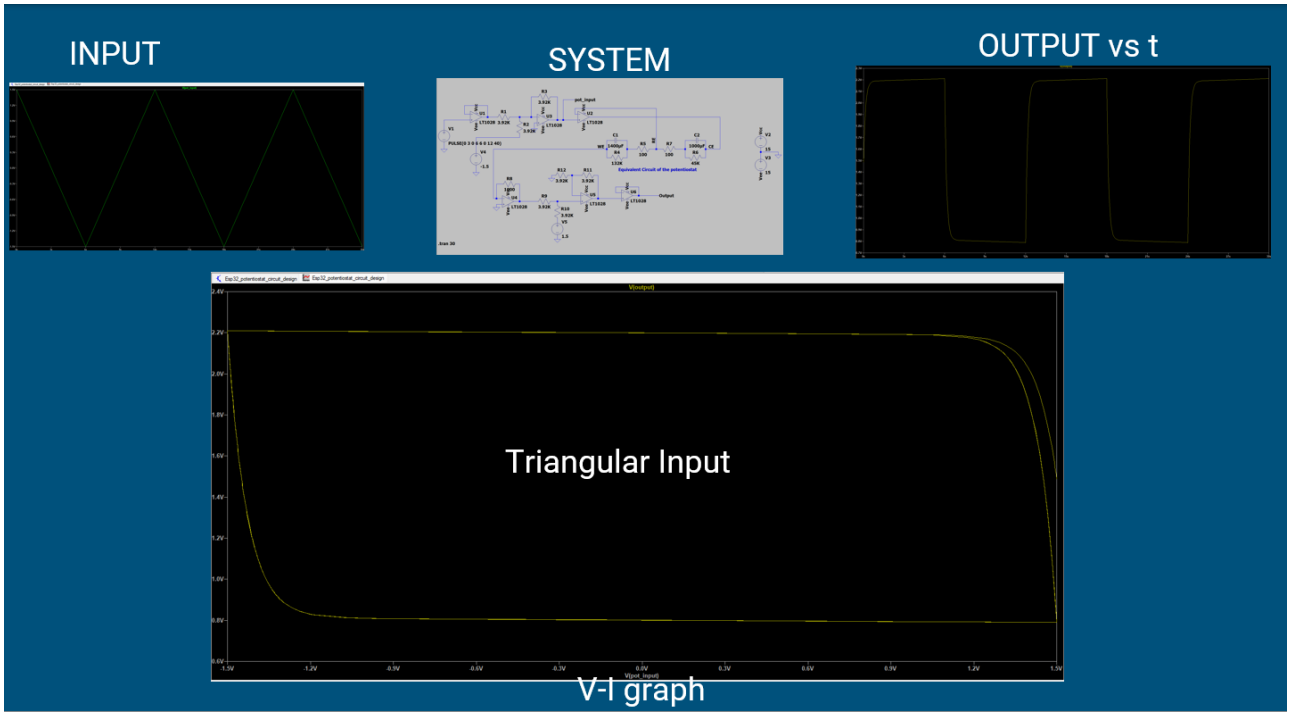
ESP32 Microcontroller - The ESP32 works as the digital control and communication unit. In LTspice, the ESP32 was represented using voltage sources to act like its DAC outputs, and voltage probes were used to measure signals.

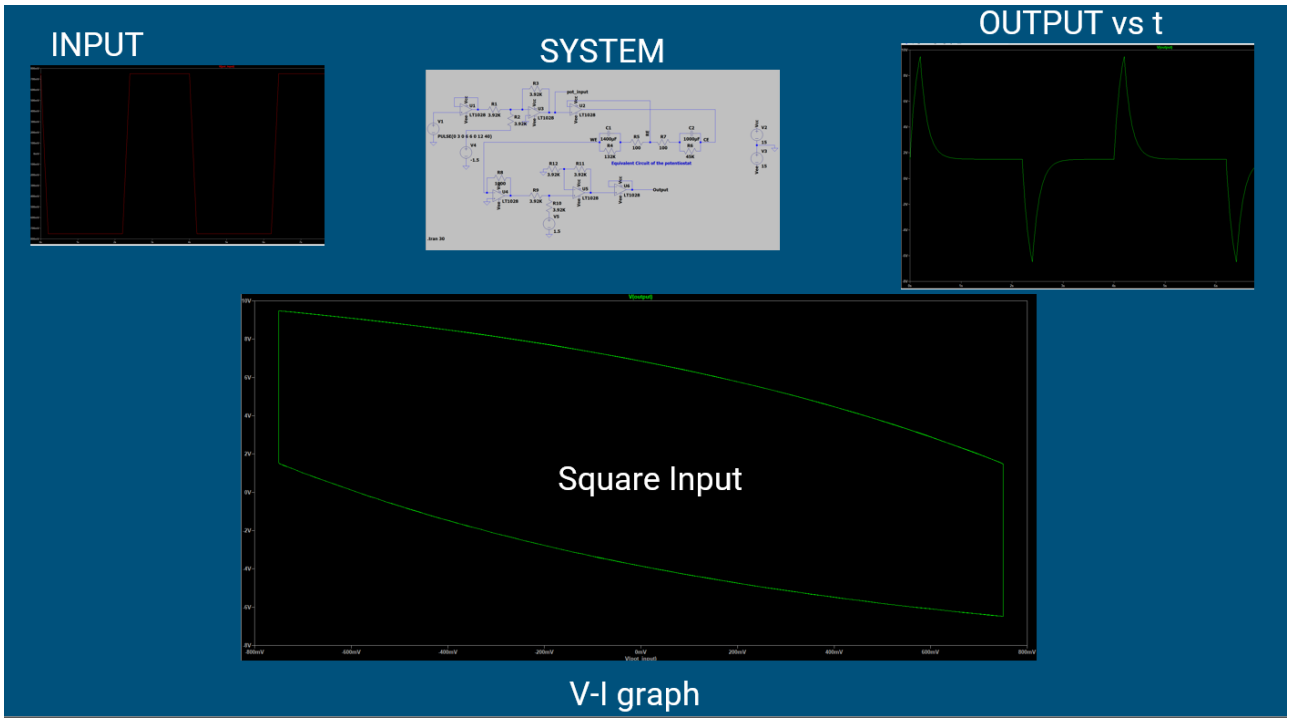
LMP91000 Analog Front-End - It interfaces directly with the Working (WE), Reference (RE), and Counter (CE) electrodes of the electrochemical cell. In LTspice, this part was modeled using an op-amp circuit with feedback resistor (Rf) to simulate the TIA function and a voltage-controlled source to emulate the cell bias control.

The initial design had several limitations, and with the guidance of my senior, Mr. Abhiram, we were able to develop the final working version shown below.

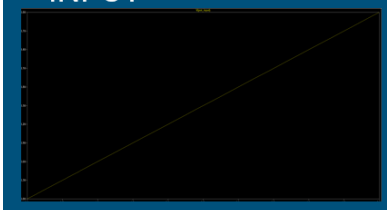


Results of some simulations

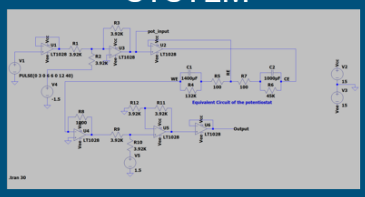




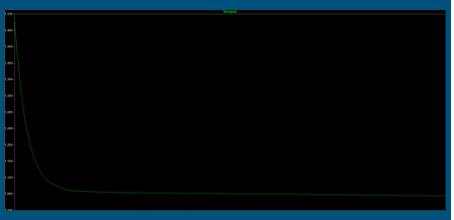
INPUT

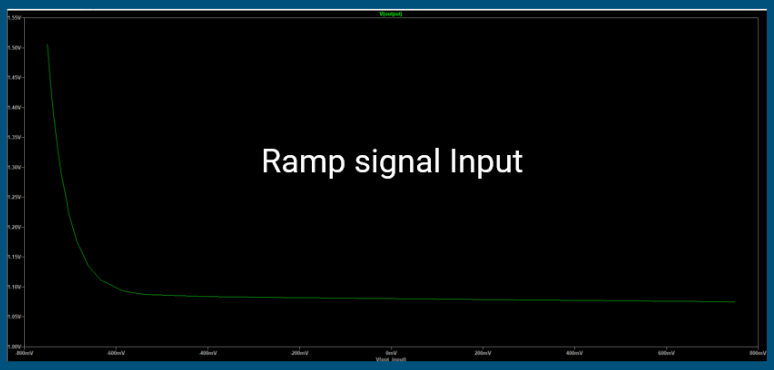


SYSTEM



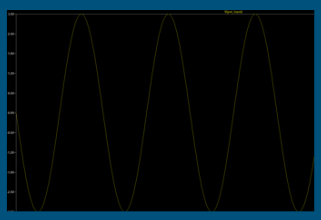
OUTPUT vs t



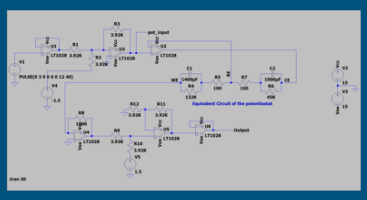


V-I graph

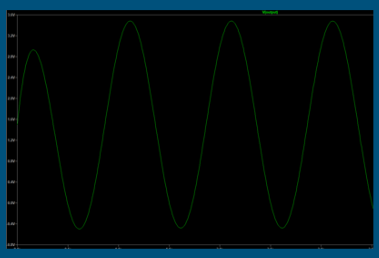
INPUT

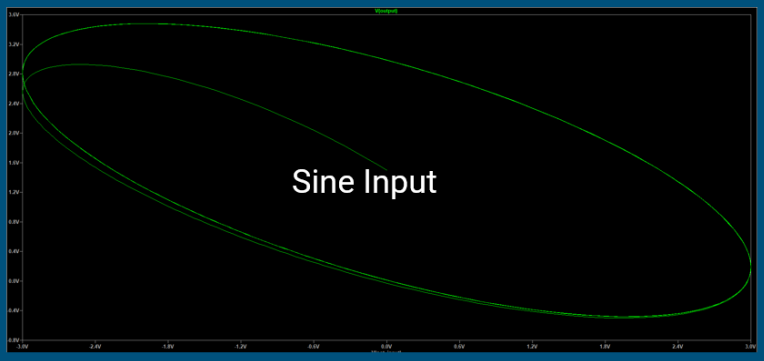


SYSTEM

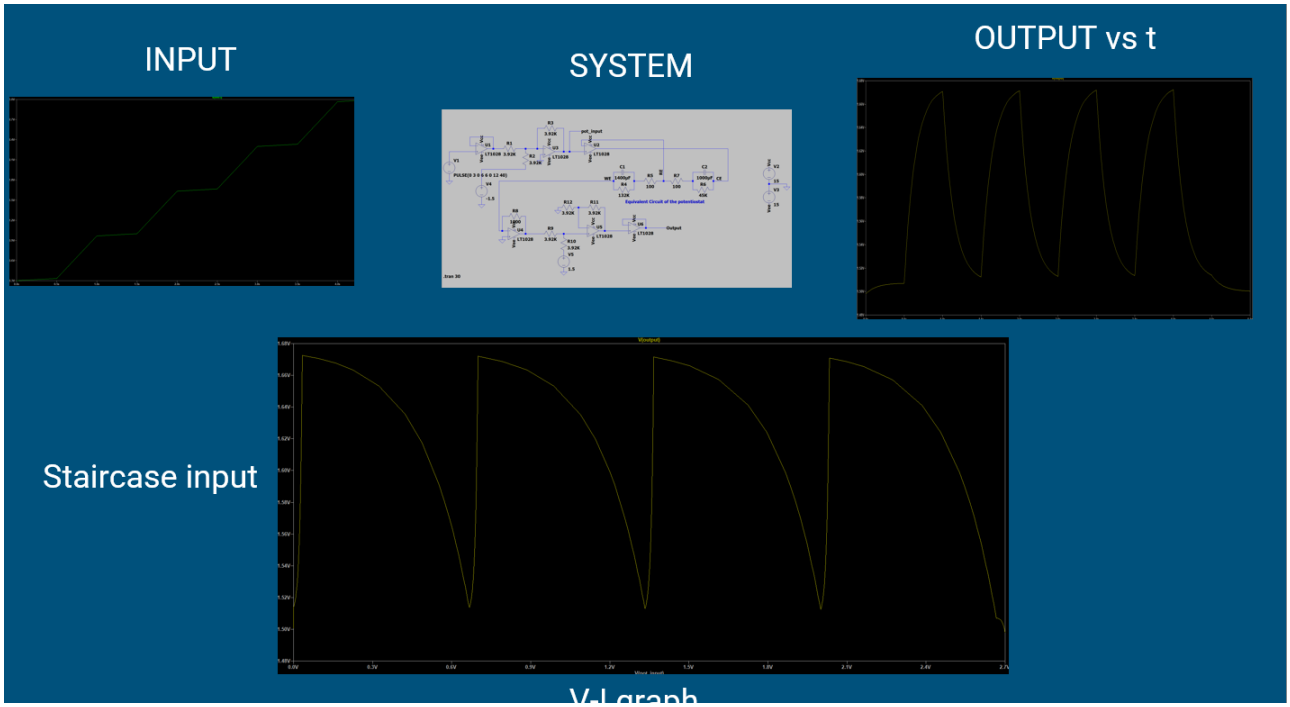


OUTPUT vs t





V-I graph



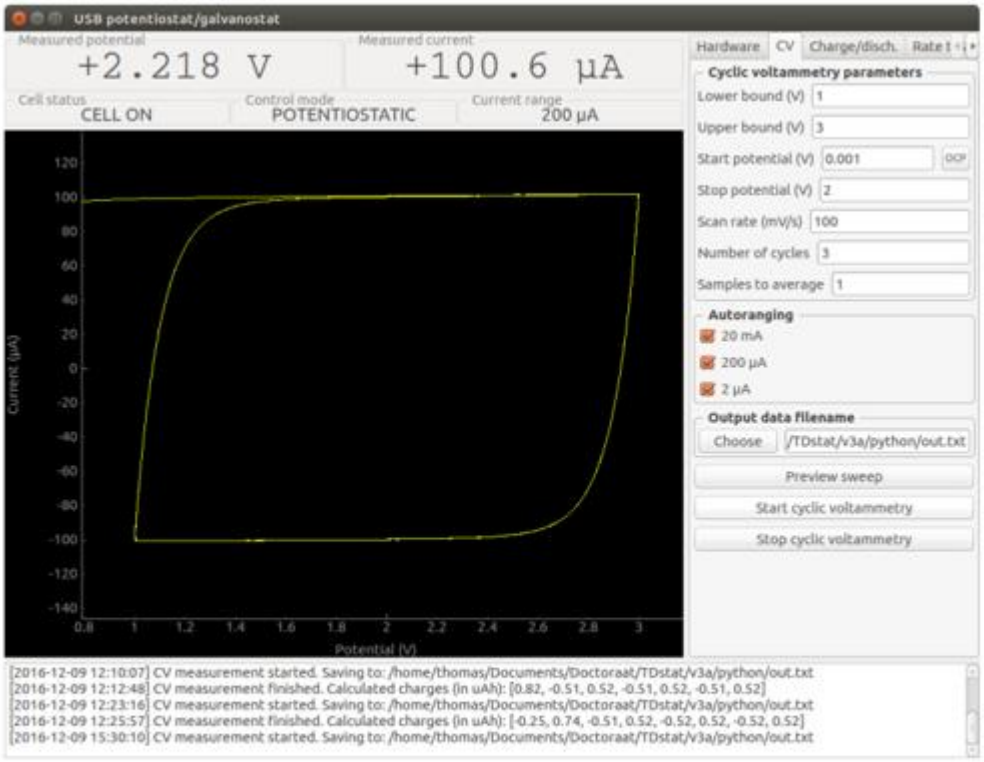
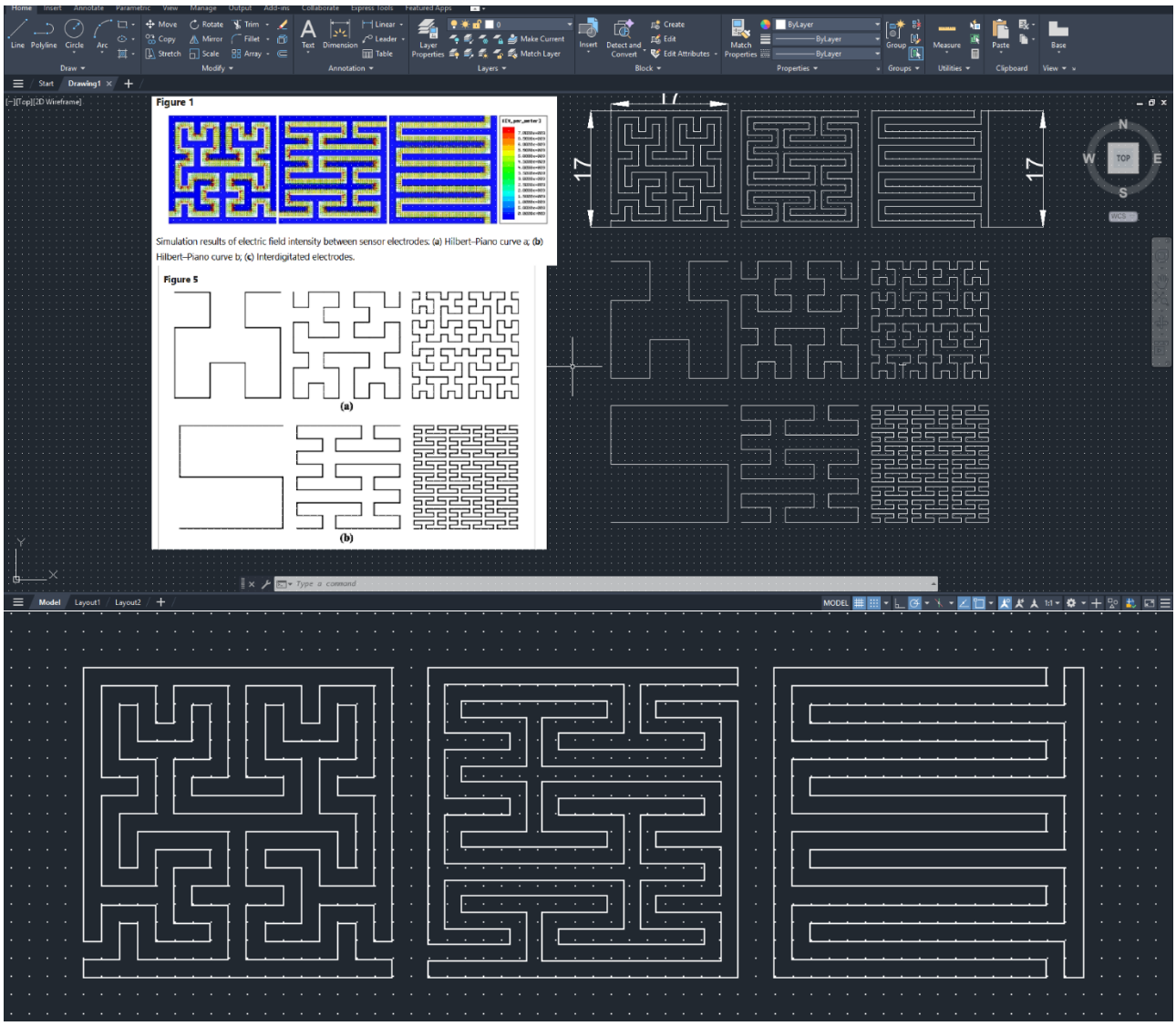


FIG. 8: Running a cyclic voltammetry measurement on the dummy cell using the "CV" tab.

Expected result is shown above

For Fractal Electrode Designs –

We started by making the electrode designs given in the paper in AutoCAD so that we can import the DXF files in COMSOL. All the electrode designs shown below are 17mmX17mm



To ensure a fair comparison, the following parameters must be kept the same for every electrode geometry:

- Minimum Gap Between Electrodes
- Total electrode area

We were able to maintain this for the 1st (Hilbert–Piano curve a) and 3rd (Interdigitated electrodes) designs as they were originally 17x17 only. The 2nd

(Hilbert–Piano curve b) design was 19x19 originally which was scaled to 17x17 which will change the electrode thickness and electrode area which further leads to inaccurate comparison. For Hilbert–Piano curve b we need to make an entirely new design with same electrode thickness and electrode area. Thus, for now we will be comparing Hilbert–Piano curve a and Interdigitated electrodes.

Common Factors Between All Electrode Designs

- Applied Voltage
- Minimum Gap Between Electrodes
- Trace Width (line thickness)
- Electrode Material
- Electrode Thickness (if 3D)
- Outer Simulation Domain Size and Shape
- Permittivity of Surrounding Medium
- Boundary Conditions (properly assigned +V and Ground)
- Mesh Refinement Strategy (same settings on all electrodes)
- Study Type and Physics Settings (DC electrostatics)
- Total electrode area

COMSOL methodology

1. **Software Setup**
    - Open COMSOL Multiphysics
    - Create a 2D model with the Electrostatics (es) physics and Stationary study
2. **Geometry Import**
    - Import electrode layouts (Hilbert A, Interdigital) in mm units
    - Draw a surrounding air domain from sketch section (rectangular region)
3. **Material Assignment**
    - Assign Air (user-defined) with $\varepsilon_r = 1$ to the outer domain
    - Electrodes kept as boundary conductors (no bulk material)
4. **Boundary Conditions**

- Apply Electric Potential (e.g., +5 V) to one electrode
- Apply Ground (0 V) to the opposite electrode
- Set outer boundary as Electric Insulation (default)

5. **Mesh Generation**

- Use coarse/normal global mesh
- Add refined boundary mesh only on electrode edges (finer mesh near corners)
- Build mesh and verify refinement visually
- A combined mesh generation approach was employed to correctly capture steep field gradient around electrodes' edges without increasing too much load on the solver. Initially, a Normal/Coarse global mesh was generated for the whole air region. Using an ultra-fine mesh in all domains led to problems such as increased computation time, unstable behavior of the solver, and poor mesh quality, which included numerous distorted and thin triangles.

  In order to solve the above problems, a mesh generation technique focusing on the boundaries of the electrodes was applied only to the electrode boundaries due to their sharp corners and narrow gaps leading to high field concentration. This way, small-sized and good quality mesh elements were provided around the critical areas while larger elements were retained in the air bulk. The same mesh was generated for both electrode configurations.



Free Triangular mesh generation algorithm was employed to automatically create triangular elements over the entire 2D region, allowing flexibility for refinement near complex geometry of the electrodes.

6. **Solve the Model**

- Stationary solver run to calculate electric potential distribution.

7. **Electric Field Evaluation**

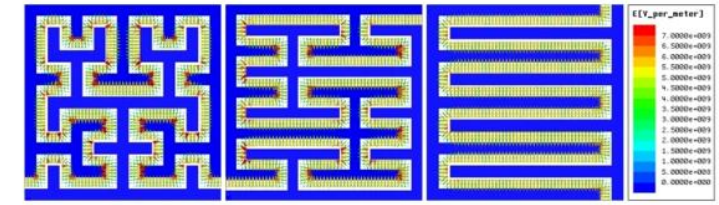
- Create a Surface Plot of es.normE to visualize electric field strength
- Extract maximum electric field using:
  - Derived Values -> Maximum -> Surface Maximum
  - Select air domain
  - Expression: es.normE

8. **Compare Results**

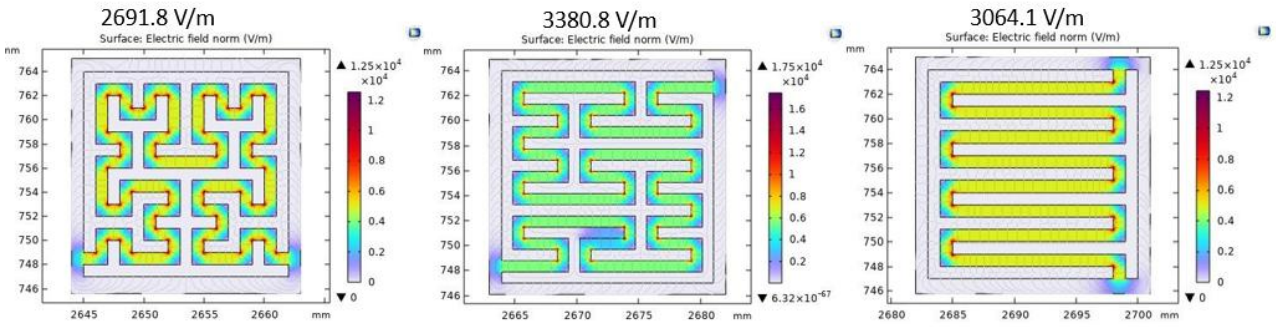
- Repeat same steps for both electrode geometries
- Compare peak electric field values to analyze performance differences

Initial results

Figure 1



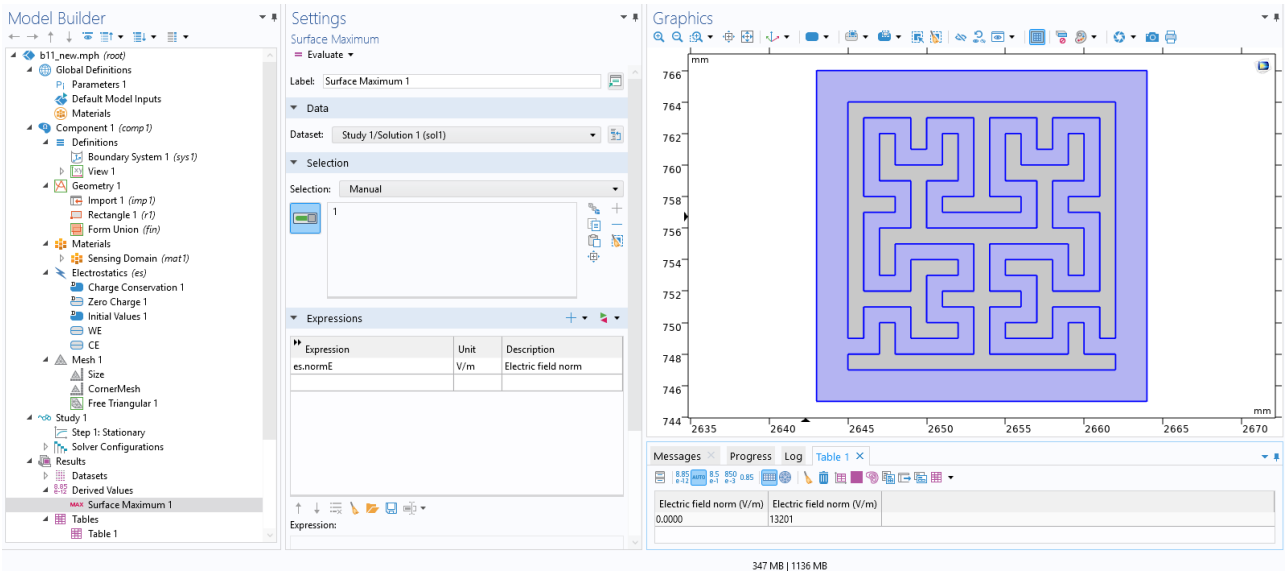
Simulation results of electric field intensity between sensor electrodes: (**a**) Hilbert–Piano curve a; (**b**) Hilbert–Piano curve b; (**c**) Interdigitated electrodes.

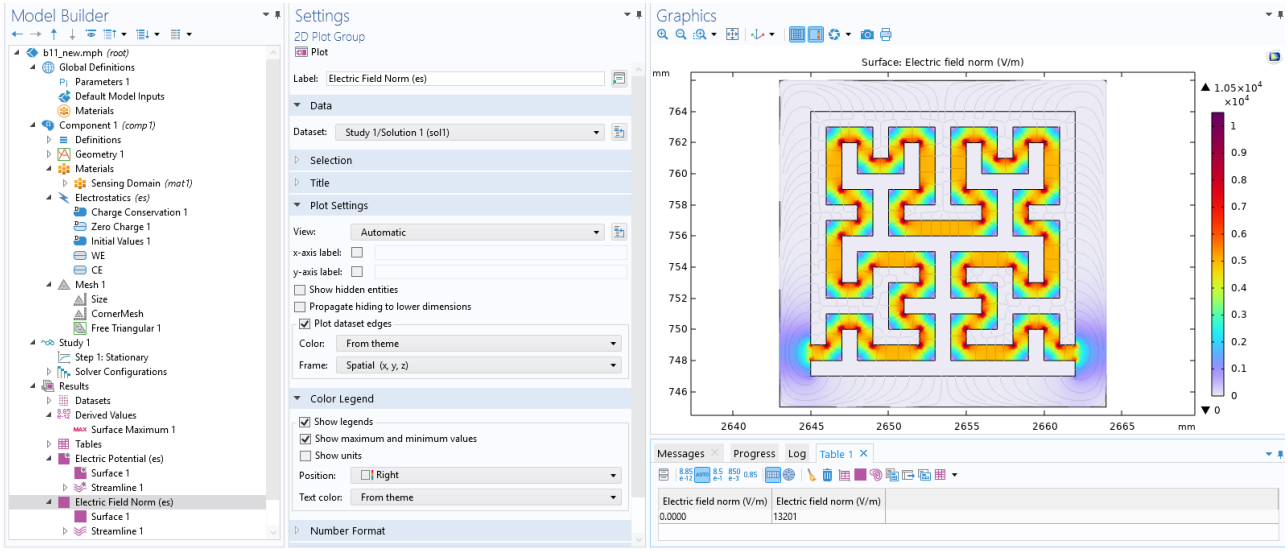


The order we get here is wrong and this is because of two reasons

1. Hilbert-Piano curve b not being constructed properly.
2. Mesh settings not being adjusted correctly.

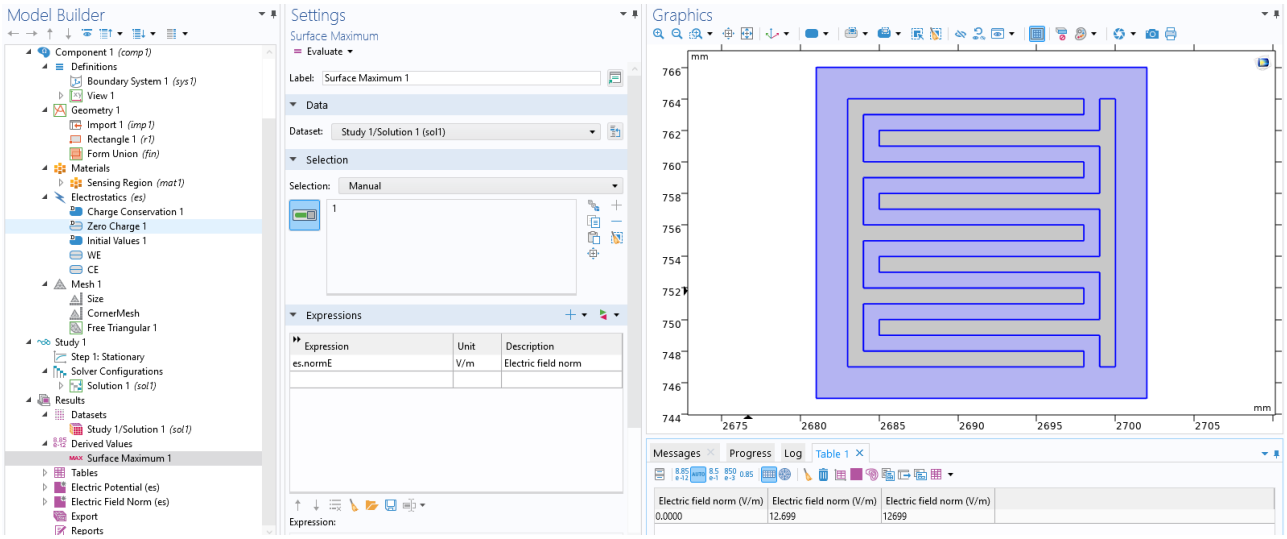
Results for Hilbert–Piano curve a

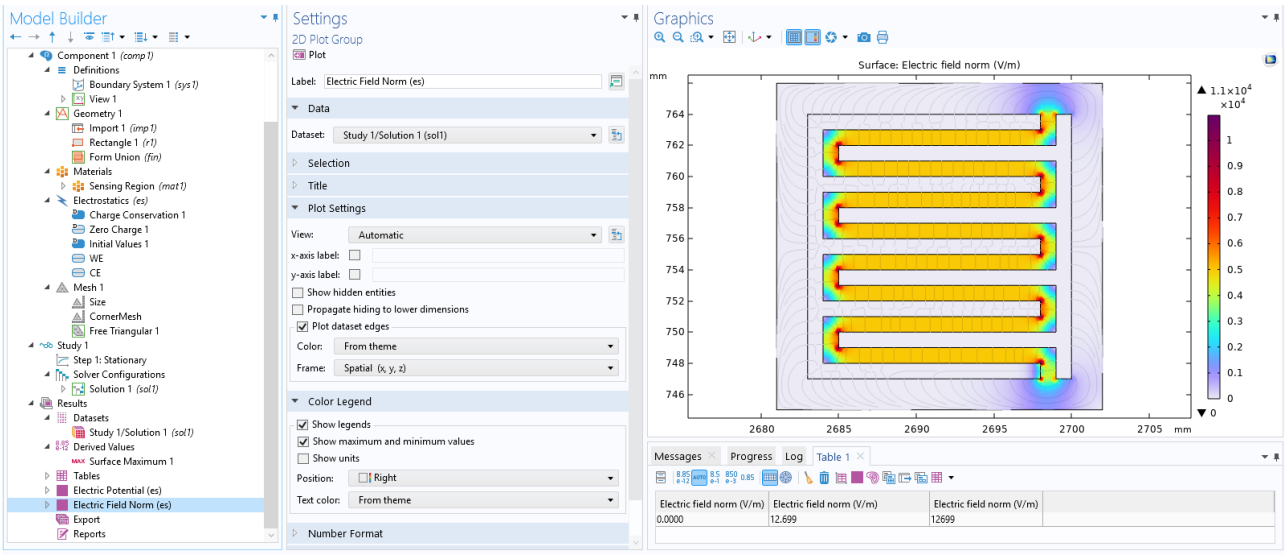




Here Electric Field = 13201V/m

Results for Interdigitated electrode





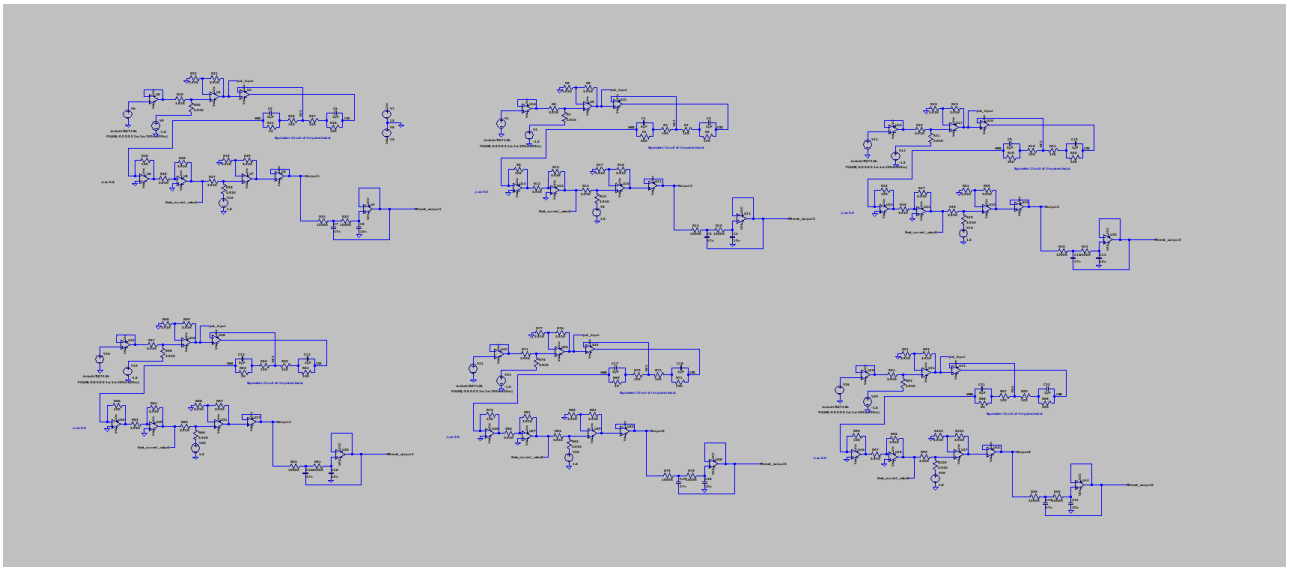
Here Electric Field = 12699V/m

This follows the expected trend which should have been E(Hilbert–Piano curve a) > E(Interdigitated electrode)

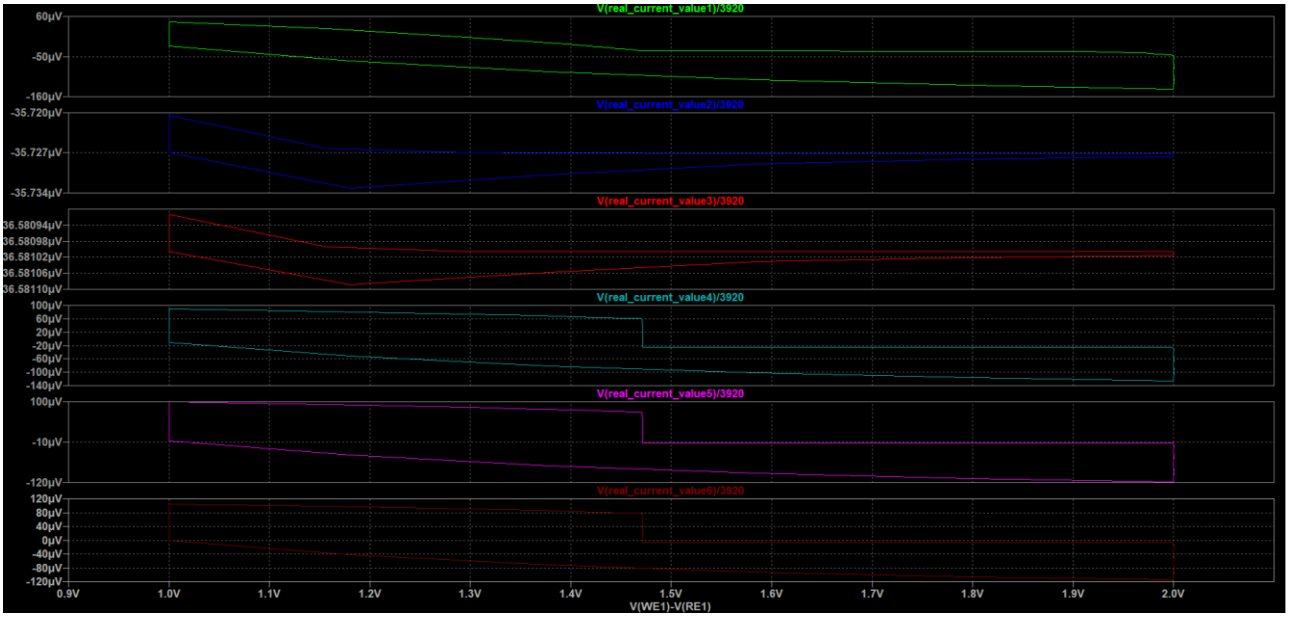
For Eprobe–

A graphical representation of the design of the six-channel potentiostat system in LTspice is presented in Figure 1, wherein each block denotes an individual potentiostat circuit. Each channel employs a controller amplifier to ensure the proper potential difference between the working electrode (WE) and the reference electrode (RE) in addition to the current required by the counter electrode (CE). For each channel, the Randles equivalent circuit is utilized to characterize the electrode–electrolyte interface. Subsequently, the electrochemical current is measured as a voltage via the transimpedance amplifier (TIA) and filtered to yield a steady output

signal. Given that the six potentiostats function independently, the proposed system allows for real-time monitoring of various electrochemical sensors at the same time.



Results of some simulations

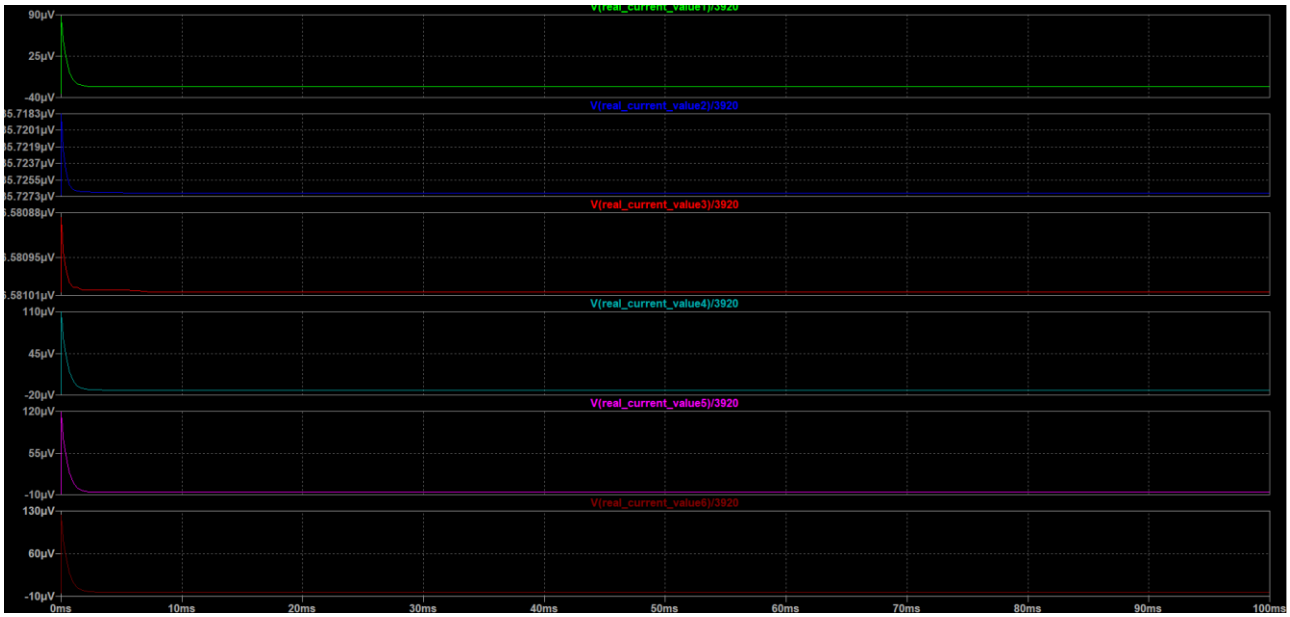


For this simulation each potentiostat has different R_{ct}

R_{ct} - connected in parallel with the double-layer capacitance C_{dl}.

R_{ct1} = 1k Ohms, R_{ct2} = 500 Ohms, R_{ct3} = 250 Ohms, R_{ct4} = 2k Ohms, R_{ct5} = 4k Ohms, R_{ct6} = 8k Ohms

The figure shows the simulated cyclic voltammetry (CV) responses obtained from the six independent potentiostat channels implemented in LTspice. The subplots denote the current generated electrochemically in one channel, determined as V(real_current_value)/3920, as a function of the electrode potential difference V(WE)-V(RE). The triangular voltage signal was employed for the scanning process of the working electrode, and the current signal obtained from it using the transimpedance amplifier is illustrated by the plots. It can be seen that at the same time, six potentiostat channels generate their current independently. Similar voltammograms' shapes are explained by the equal model parameters, while some differences illustrate independent channels' operation.



The chronoamperometry (CA) simulation shows the current response of the electrochemical cell after a step change in the applied potential. A sharp current peak appears immediately after the voltage step due to rapid charging of the electrode double-layer capacitance, followed by a gradual decay as diffusion limits the electrochemical reaction. The results demonstrate that the potentiostat circuit is capable of measuring transient electrochemical currents and performing chronoamperometric analysis.

For Two-Channel potentiostat-

After this, we first started with Two-channel potentiostat for simplicity after which we can move forward to multi-channel potentiostats.

For this we first started with two different potentiostats which needed to be connected and made compact. For this a summing amplifier circuit can work based on the reference literature. The circuit should be effectively resembling the image given below -

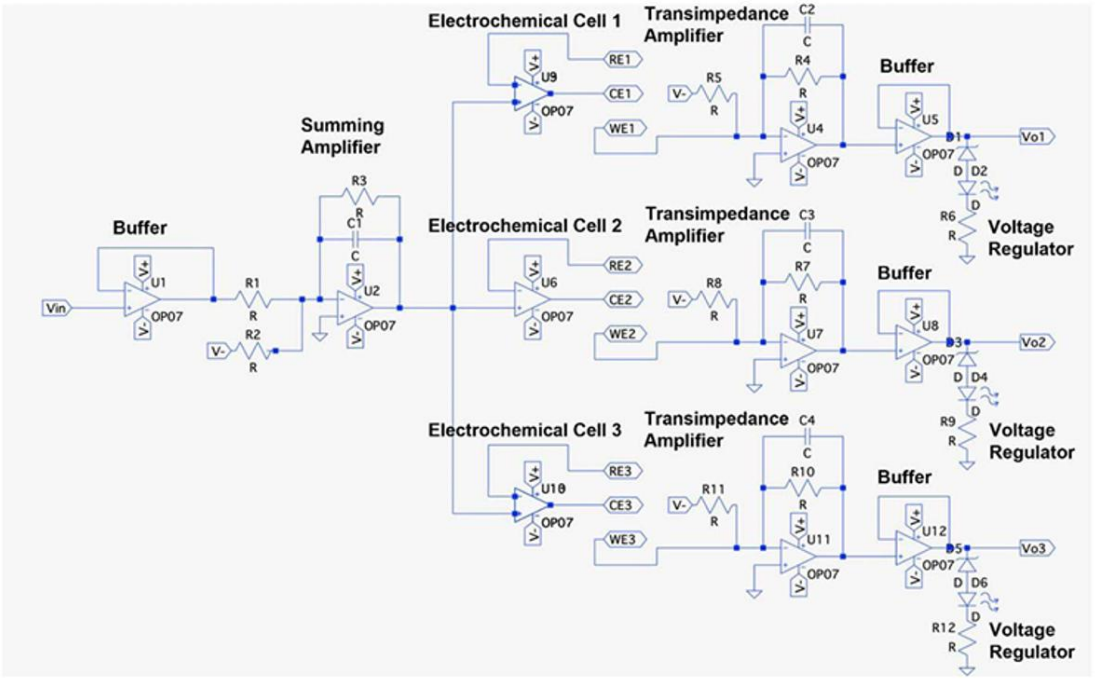
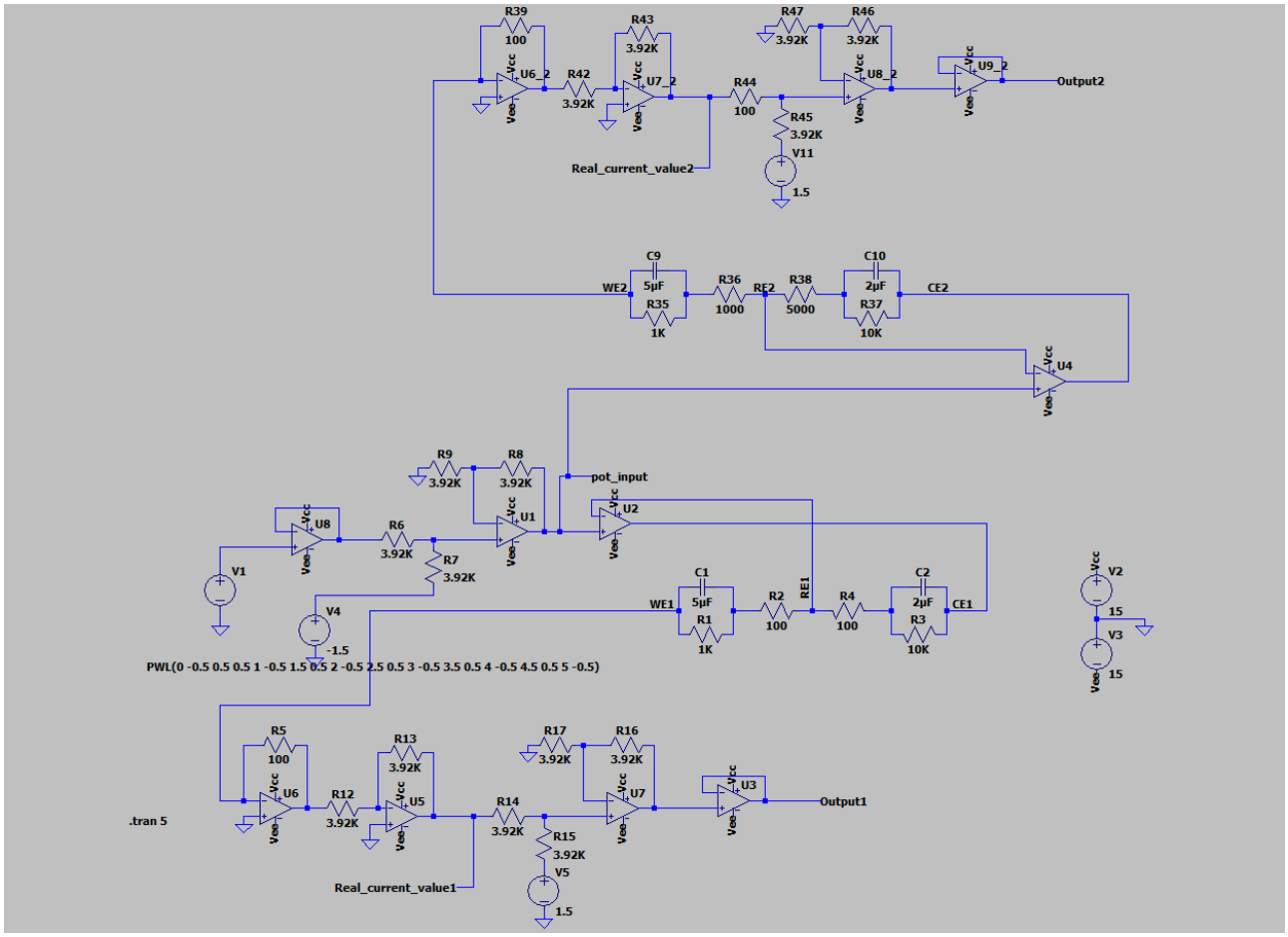


FIGURE 2. Schematic design of the analog front-end circuit of ESPotensio.

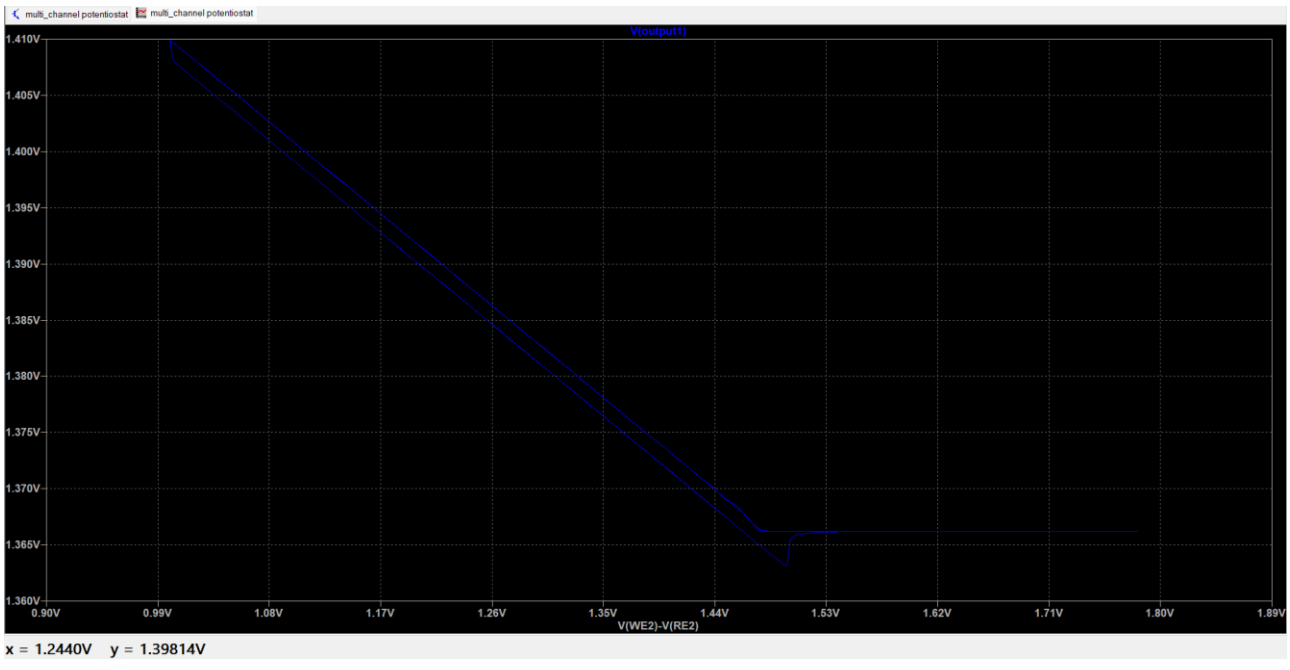
The summing amplifier forms an integral component of the control voltage generation stage in the potentiostat, through its ability to add the input excitation signal to a reference voltage. Its function here is to make sure that the sweep voltage is applied simultaneously and consistently to all the channels, thus making it an indispensable component when carrying out multi-channel electrochemical analysis. The resistance network is used in setting the gain and the weightage of each input channel, thereby enabling the control of output bias and gain. The addition of the feedback capacitor is intended to stabilize the amplifier operation and reduce high-frequency noise. The summing amplifier had low output impedance, which makes it suitable for driving the various channels without any degradation of signals.

The first circuit made was as given in the image below

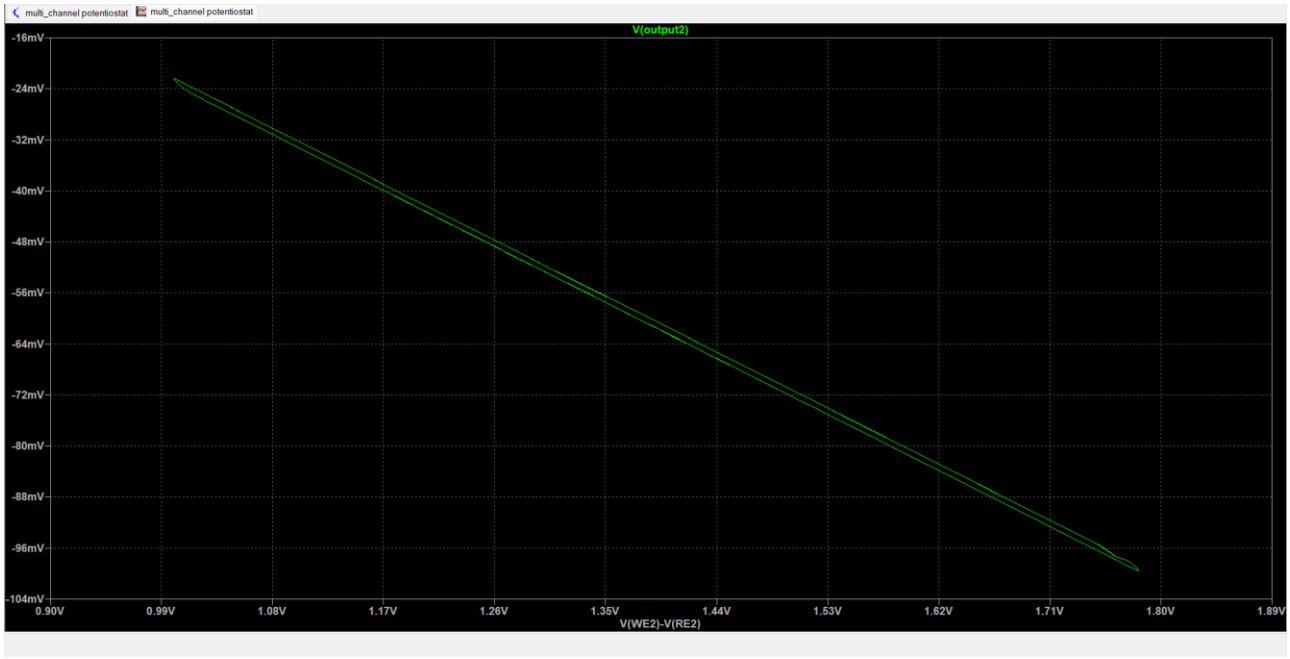


In this circuit the Randles' equivalent cells were used as inputs.

The outputs that we got were

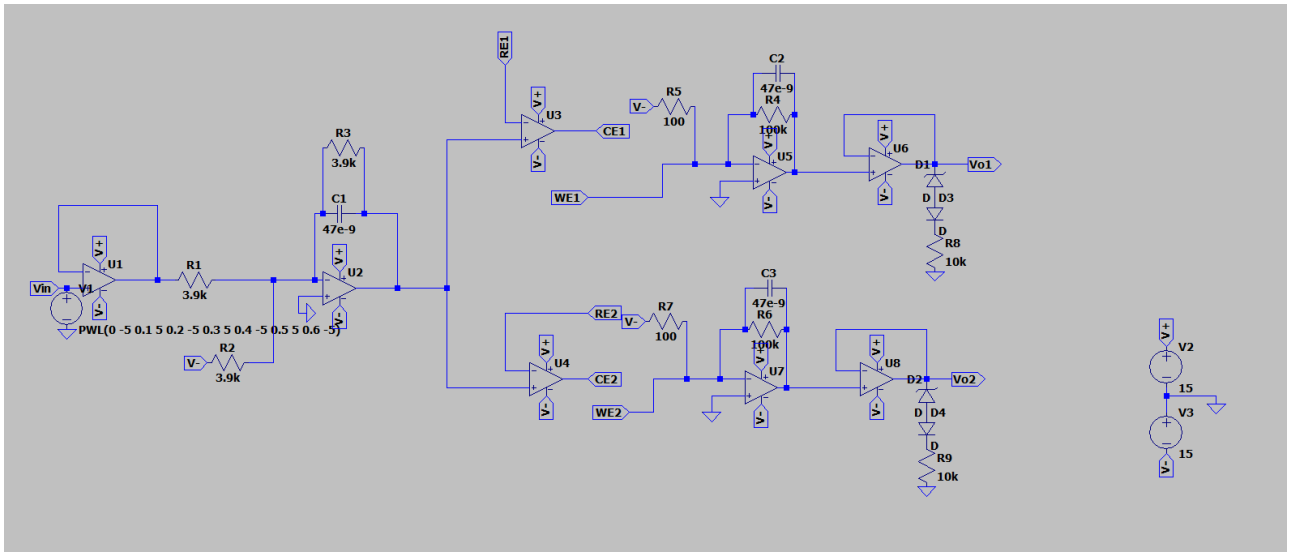


Current at Output 1 vs V(RE1) – V(WE1)

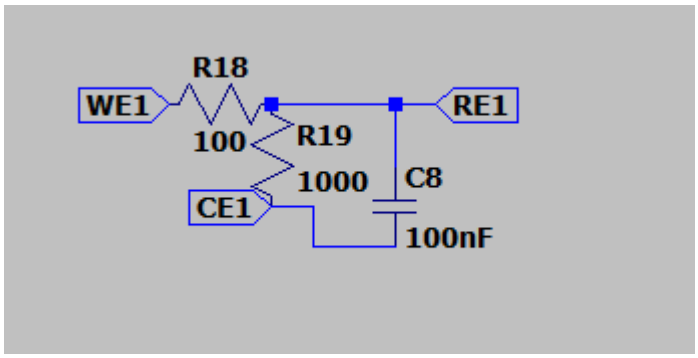


Current at Output 2 vs V(RE2) – V(WE2)

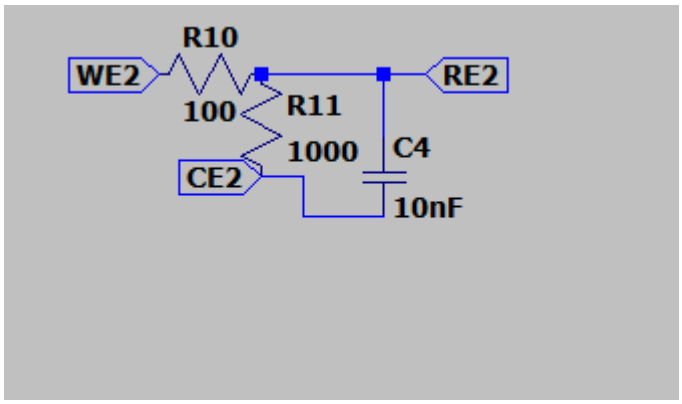
The output response of the system is very dependent on the internal and intrinsic characteristics of the electrochemical cell, such as its impedance, reaction kinetics, and capacitance. Therefore, the output waves produced by different cells would be greatly affected despite being exposed to similar exciting stimuli. During the preliminary simulations and analysis, it became evident that the circuit implemented suffered from lack of clear and consistent behavior, resulting in substandard outcomes. Therefore, to enhance the control and consistency, it was decided to alter the design methodology to improve its efficacy. The new design is presented in figure x, offering improved results compared to the old one.



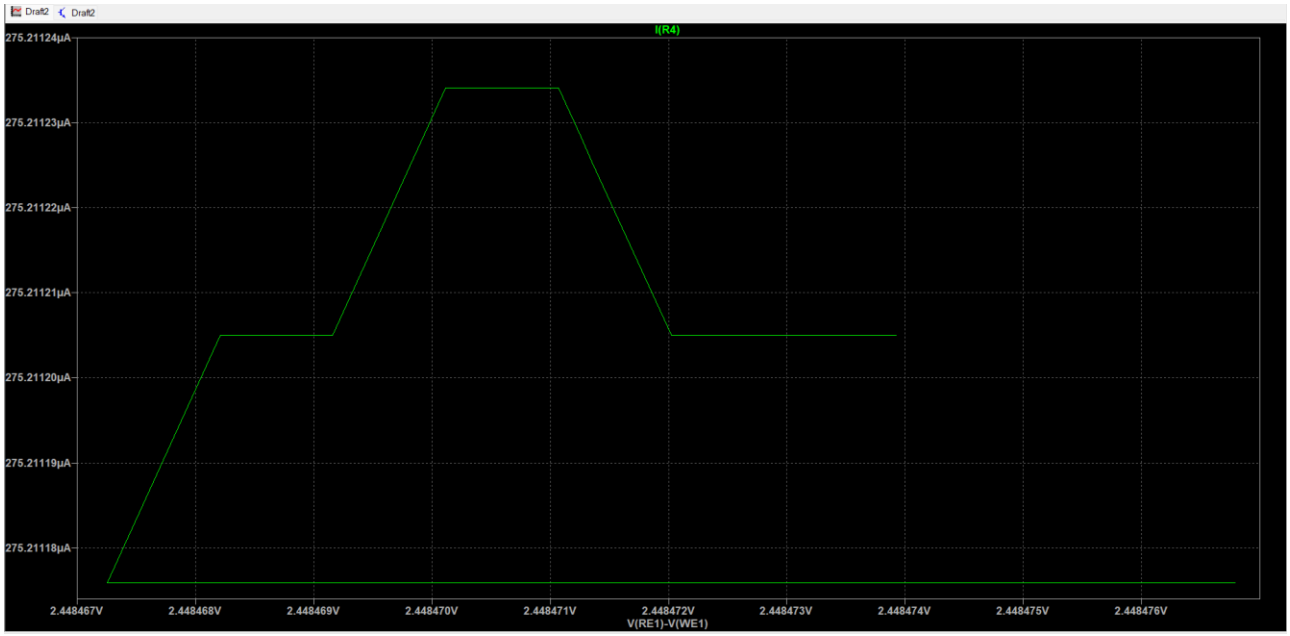
The potentiostat circuit



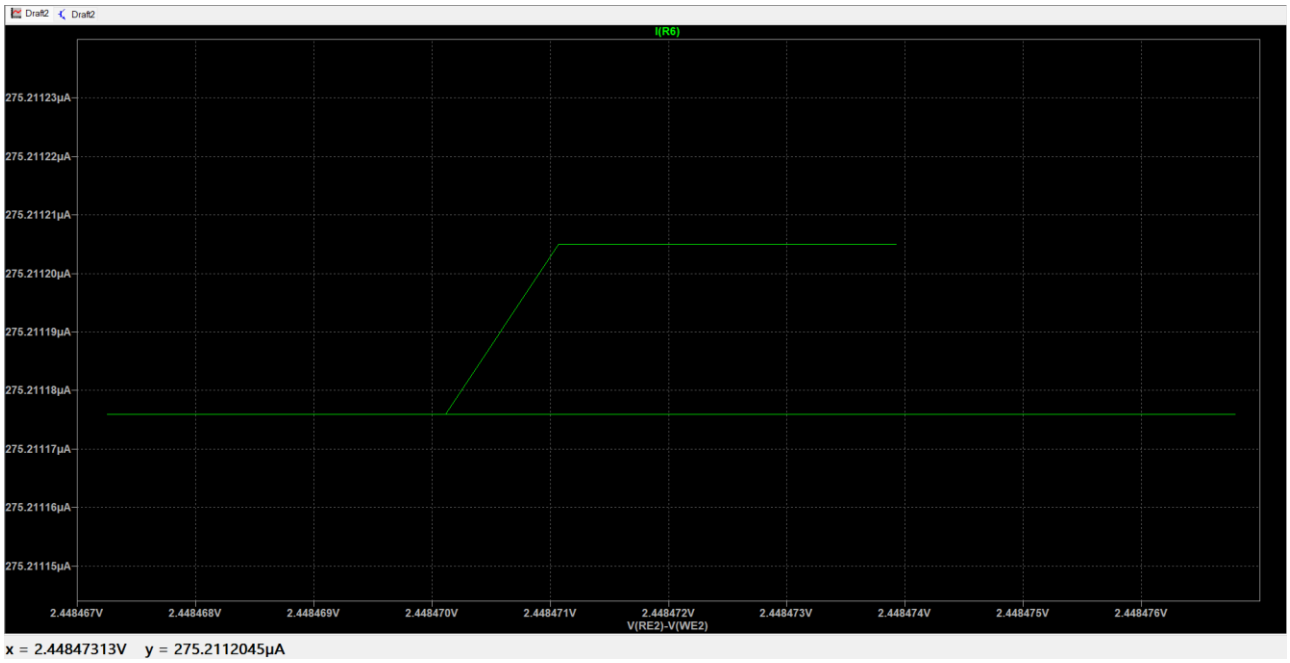
Equivalent cell for first circuit



Equivalent cell for first circuit



Output 1(V01) when input is a triangular wave



Output 1(V02) when input is a triangular wave

The constructed circuit constitutes a multi-channel potentiostat where a triangular signal is processed using a buffer and a summing amplifier to produce the control voltage to be supplied to the electrochemical cell. In each channel, there is a control amplifier responsible for maintaining the desired potential difference between the working and reference electrodes, the Randles equivalent circuit to model the electrochemical interface (consisting of solution resistance, charge transfer resistance, and capacitance of the double layer) in addition to the transimpedance

amplifier used to convert the produced current into a voltage. The output buffer plays an essential role in stabilizing and protecting the signal. The simulated current-voltage relationship indicates that the system operates within a highly resistive mode because of its extremely linear characteristic without much fluctuation. This results from the small voltage sweep seen across both the working and reference electrodes. This causes the absence of any significant effect due to the lack of dynamic changes. Moreover, the absence of a CV loop can also be attributed to that fact.

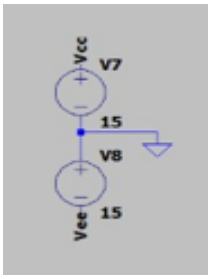
The next step would be to improve this output by changing the required parameters and then implement this circuit on a breadboard using real components.

CHAPTER 6 - Elements Of Potentiostat

1. Power Supply and Setup

The op-amps (TL074) are powered by ±15 V from sources V7 (+15 V) and V8 (–15 V).

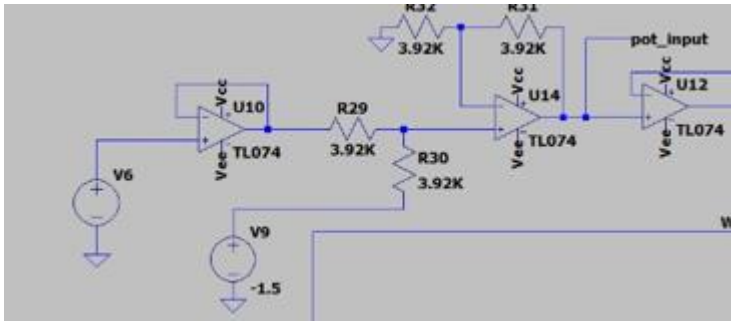
TL074 is chosen because it's a low-noise, JFET-input op-amp, ideal for precise analog circuits like potentiostats.



2. Signal Input Section (Top Left)

V6 and V9 generate the input bias or reference potentials for the control loop.

U10, U12 and U14 (TL074 stages) condition this input control signal (pot_input) usually a triangular or step waveform representing the applied voltage (for cyclic voltammetry).



3. Electrochemical Cell Model (Center)

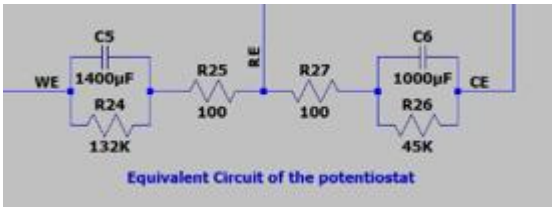
This is the main part of our simulation:

- R24, C5, R25, R27, C6, R26 model the electrochemical cell — including:
  - Solution resistance (R25, R27)

- Double-layer capacitance (C5, C6)
- Faradaic resistance or charge-transfer resistance (R24, R26)

The three terminals:

- WE (Working Electrode) -- where the potential is controlled.
- RE (Reference Electrode) -- provides a fixed potential reference.
- CE (Counter Electrode) -- completes the circuit and supplies current.

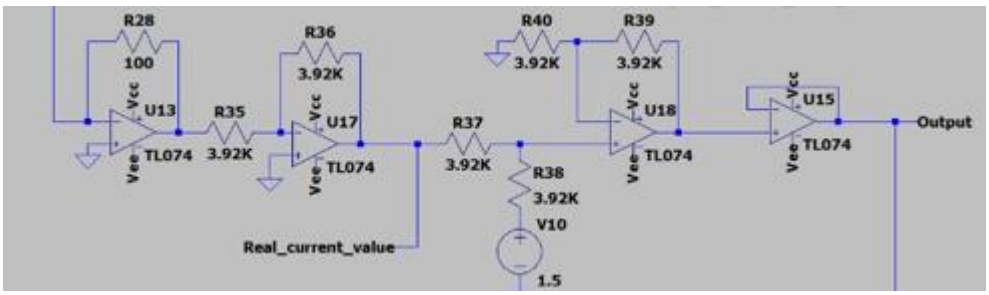


4. Feedback and Control Section (Middle & Bottom Left)

- The op-amps U13, U17, U18, U15 form the control loop of the potentiostat.
- The goal is to maintain the WE–RE voltage constant (the "set potential") by adjusting the CE potential through feedback.

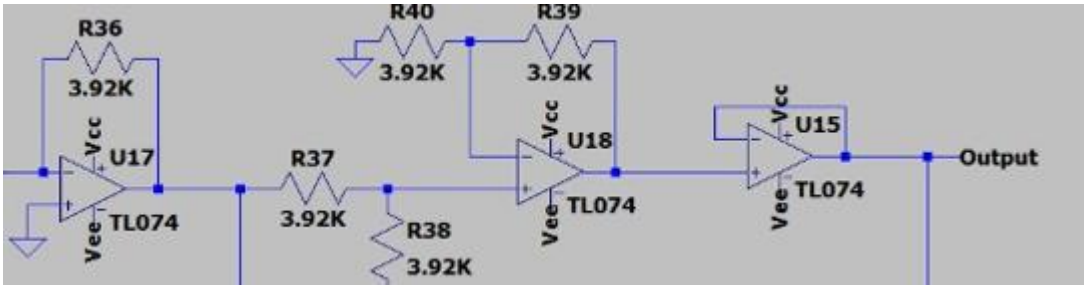
How it works:

1. The input waveform (pot_input) sets the desired WE–RE voltage.
2. The potentiostat senses the actual WE–RE voltage.
3. The feedback op-amps adjust the current through CE to correct any error.
4. The output current is proportional to the electrochemical reaction current.

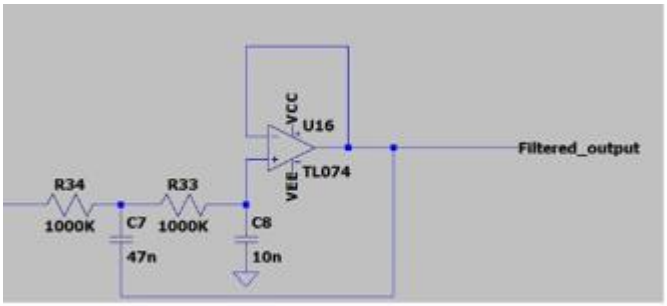


5. Measurement Path

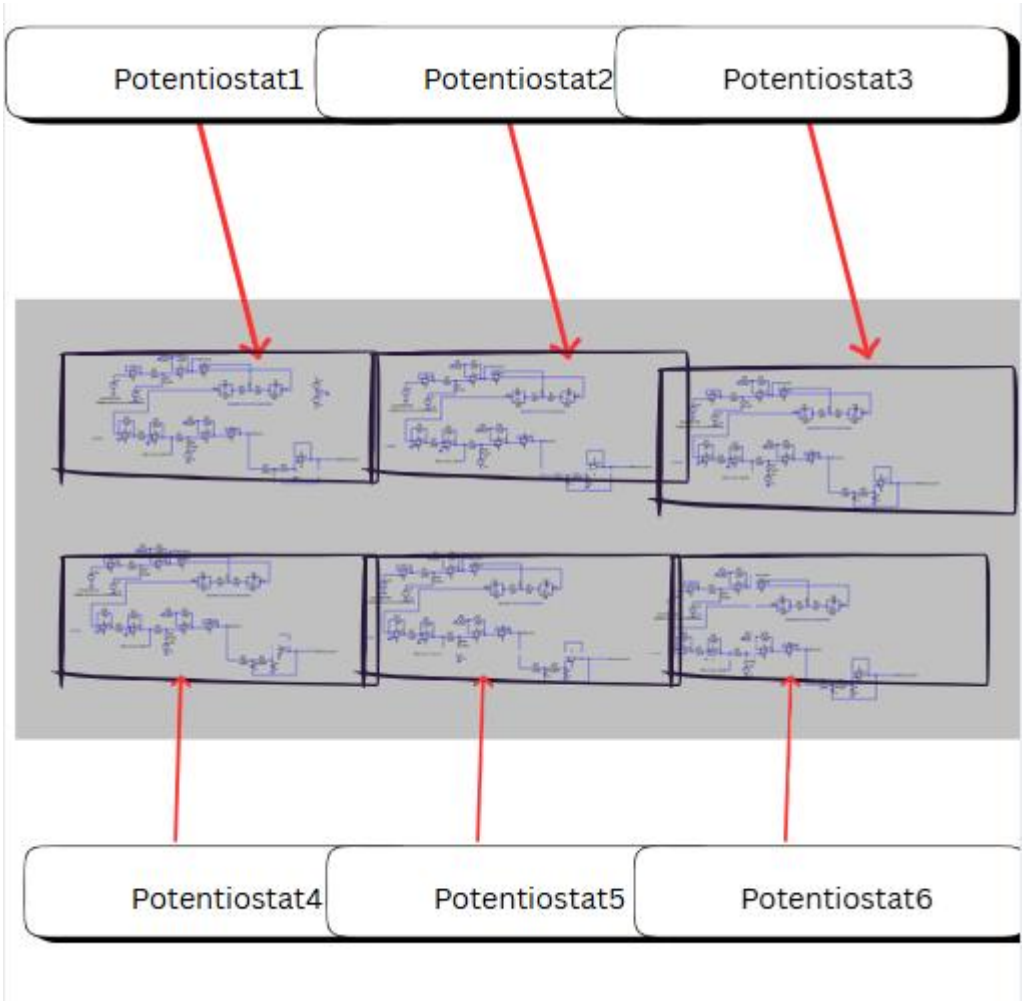
- “Real_current_value” node measures the current flowing through the working electrode .
- U15 and resistors R37–R40 amplify and convert this current to a measurable voltage output (Output).
- This represents our I–V (current vs. potential) curve signal.



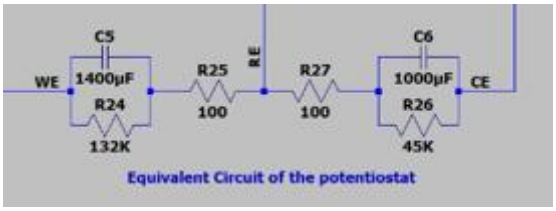
6. **Output Filtering Section (Right Bottom)**
   - U16, R33, R34, C7, C8 form a low-pass filter.
   - This removes high-frequency noise, leaving a clean Filtered_output signal which is ideal for data recording or plotting cyclic voltammograms.



CHAPTER 7 - Elements Of Eprobe



Randles Equivalent Circuit



Each potentiostat behaves independently while having its own working, reference and counter electrode along with its own Randles Equivalent circuit. The power supply is same for each potentiostat as the op-amps should connect to the same rails.

Each circuit will give its own unit output and filtered output. The input of each potentiostat being same is a common configuration and if the potentiostat parameters are different, we will get different outputs.

1. **Analog Front-End (AFE)**
   The analog front-end helps control the electrode potential and measures the electrochemical current. In our LTspice design, this is implemented using operational amplifier stages (UniversalOpamp2) that form the potentiostat control loop and transimpedance amplifier. These units regulate the potential between the working electrode (WE) and reference electrode (RE) and convert the measured current into a voltage signal.

2. **Electrochemical Cell Interface**
   The electrochemical interface represents the three electrode system consisting of WE, RE, and CE. In our design, this interface is made using the Randles equivalent circuit, which includes components such as solution resistance (R_s), charge transfer resistance (R_{ct}), and double-layer capacitance (C_{dl}).

3. **Transimpedance Amplifier (Current Measurement Unit)**
   The transimpedance amplifier takes the electrochemical current produced by the working electrode and converts it into a voltage signal. In our design, the op-amp with feedback resistance (for instance, R36 = 3.92 kΩ) acts as the transimpedance amplifier based on the formula

$$V_{out} = -I_{cell}R_f$$

4. **Signal Conditioning and Filtering Unit**
   The next stage is the filtering process where the converted signal passes through filters and amplifiers to eliminate the noise. The filters used in our experiment include resistor-capacitor filters and operational amplifiers to filter out the measured signal.

5. **Multi-Channel Measurement Architecture**
   The system also helps to integrate six identical potentiostat channels in our simulation. Each channel contains the control amplifier, electrochemical interface, and current measurement stage, allowing simultaneous electrochemical measurements from multiple sensors.

6. **Input Waveform Generator**
   The input voltage source provides excitation waveforms used for electrochemical techniques. In our LTspice circuit, this is implemented using voltage sources (e.g., V6, V7, etc.) that generate waveforms such as triangular signals for CV, ramps for LSV, or step voltages for CA.

CHAPTER 8 - Elements Of Two-Channel Potentiostat

1. **Input Voltage Source (Vin)**
   Generates the excitation signal (triangular/PWL waveform) and helps provide the voltage sweep required for cyclic voltammetry.

2. **Buffer Amplifier (U1)**
   Unity gain op-amp configuration, prevents loading of the input source and helps provide low output impedance.

3. **Summing Amplifier (U2)**
   Combines input signal with reference voltage, generates the final control voltage (pot_input) while the resistors (R1, R2, R3) set gain and offset and capacitor (C1) stabilizes the circuit.

4. **Control Amplifiers (U3, U4)**
   Drive the counter electrodes (CE1, CE2) and ensures $V(WE) - V(RE) = V_{input.}$

5. **Electrochemical Cell Model (Randles Circuit)**
   Each channel includes:
   - Solution Resistance (R_s) – R18, R10 etc.
   - Charge Transfer Resistance (R_{ct}) – R19, R11 etc.
   - Double Layer Capacitance (C_{dl}) - C8, C4 etc.
   - Electrodes
     - WE (Working Electrode) - where current is measured
     - RE (Reference Electrode) - sensing node
     - CE (Counter Electrode) - driven by op-amp

6. **Transimpedance Amplifier (TIA) (U5, U7)**
   Converts current to voltage.
   Components:
   - Feedback resistor (R4, R6 = 100kΩ)
   - Feedback capacitor (C2, C3 = 47nF)

   Relation:

$$V_{out} = -I_{WE} \times R_f$$

7. **Bias / Input Resistors (R5, R7)**
   Provides bias path for op-amp inputs and improves stability and prevent floating nodes.

8. **Output Buffer (U6, U8)**
   Isolates TIA from load, provides stable output voltage and helps ensures low output impedance.

9. **Protection Diodes (D1–D4)**
   Clamps output voltage and protect circuit from overvoltage.

10. **Output Load / Regulator Resistors (R8, R9)**
    Limits current through diodes and provide reference path to ground.

11. **Power Supply (V2, V3)**
    Dual supply: +15V and −15V

12. **Net Labels (WE1, RE1, CE1, WE2, RE2, CE2)**
    Defines electrode connections, ensures channels remain isolated and prevents unintended short circuits.

CHAPTER 9 - Some Real-Life Applications

- An electrochemical cell can be used to conduct studies related to chemical analysis. For example, it can be used in oxidation-reduction reactions using techniques such as cyclic voltammetry to analyze compounds.
- It is also used in biomedical devices such as glucose meters where it detects the small current generated during the biochemical reaction for disease diagnosis purposes.
- Electrochemical cells find application in the development of environmental sensors that help detect harmful gases such as carbon monoxide and nitrogen dioxide in the atmosphere.
- It can be used in battery and fuel cells testing for charge/discharge cycles while controlling and measuring the required electrode potentials accurately.
- The electrochemical cell finds wide applications in industrial processes such as metal electroplating where it is required to precisely maintain certain electrode potentials.
- Electrochemical cells are frequently used in educational institutions such as universities and colleges as an effective teaching tool..
- Practical examples of the use of multi-channel potentiostats, such as Eprobe, include the use of electrochemical biosensors for measuring glucose levels in diabetic individuals. In such electrochemical sensors, an electrochemical sensor is employed which has enzymes, including glucose oxidase, that interact with glucose molecules. This interaction causes a redox reaction, leading to the generation of current. The amount of current generated is directly proportional to the glucose level. In addition, with a multi-channel potentiostat, one can measure from multiple sensors simultaneously to measure multiple biomarkers.

CHAPTER 10 - Advantages and Disadvantages

Advantages:

- **Accurate Potential Control:**
  It keeps a constant voltage between the two electrodes.
- **Real-Time Current Measurement:**
  Continuous measurement of the current during the electrochemical reaction is possible using the circuit.
- **Low Component Cost:**
  This circuit has been developed using cheap components including op-amps and resistors.

Disadvantages:

- **Noise and Drift Issues:**
  Analog circuits are sensitive to thermal noise, offset drift, and instability due to environmental changes over time.
- **Requirement of Shielding:**
  Measurement of the nanoamperes level of current requires either special equipment or shielding from noise.
- **Susceptibility to Environmental Factors:**
  Temperature, humidity, and EMI can influence the stability of the measurement.

For Eprobe:

Advantages

1. **Simultaneous multi-sensor measurement:**

   The use of a multi-channel potentiostat can be advantageous to measure several electrochemical sensors simultaneously, making it possible to measure multiple analytes.

2. **Higher efficiency and throughput:**

Concurrent execution helps cut down on experiment time and makes it possible to carry out real-time measurements of multiple samples and processes.

Disadvantages:

1. **Increased circuit complexity:**

   This entails greater design difficulties since more components must be integrated into the circuit design.

2. **Potential channel interference and noise:**

   Interference and noises can occur within the channels due to the lack of proper isolation and filtering.

Conclusion

Simulation and design of two-channel potentiostat were conducted using LTspice software. This was done based on a systematic approach, which included input conditioning, summing amplifier, control amplifier, and transimpedance amplifiers (TIA). The Randles equivalent cell model was used to simulate the behavior of individual cells within the circuit. The multi-channel potentiostat could control several channels at a time, although a single input excitation signal was used. Transimpedance amplifier was employed to convert the current generated by the electrodes into voltage form. Initially, there were problems in terms of the wrong implementation of the circuit and its parameters. The system performed stably after some adjustments. In addition, the output showed that current was produced. Nevertheless, the resulting waveform exhibited linear nature due to small voltage sweeps and dynamics of the system. Further tuning of both input signal and circuit parameters should be done to achieve appropriate cyclic voltammetry curve. Generally, the developed multi-channel potentiostat served as a good basis for further developments.

In terms of future development, there are possibilities to improve the circuit design in such a way that it will become more accurate, compact, and efficient. Moreover, our plan involves constructing our circuit using the real components on the breadboard, which will give us comparable results. Another point in regard to future development concerns increasing the number of potentiostat channels up to eight channels in total, so that we can use other sensors. Future developments will include nonlinearity, noise reduction, proper component value selection, and practical application of the structure.

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I. Anshori, I. F. Ramadhan, E. Ariasena, R. Siburian, J. Afif, M. Handayani, H. Yunkins, T. Kuji, T. Latifah Erwati Rajab Mengko, and S. Harimurti, "ESPotensio: A Low-Cost and Portable Potentiostat With Multi-Channel and Multi-Analysis Electrochemical Measurements," IEEE Access, 2022.